

POSITIVE ALLOSTERIC MODULATION OF THE METABOTROPIC GLUTAMATE RECEPTOR 5

Weronika CHOJNACKA¹, Fanny MALHAIRE¹, Rym BEN BOUBAKER¹, Damien MAUREL¹, Mehdi SEKSSAOUI¹, Eaazhisai KANDIAH², Cyril GOUDET¹, Xiaojing CONG¹, Carine BECAMEL¹, Guillaume LEBON¹

1. IGF, Université de Montpellier, CNRS, Inserm, 34094 Montpellier, France

2. ESRF, CM01 beamline, 38000 Grenoble, France

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ABSTRACT

The metabotropic glutamate receptor 5, mGlu₅, is widely expressed in the brain where it modulates synaptic plasticity and plays important roles in physiological and pathological processes, such as learning, memory and schizophrenia. Upregulation of mGlu₅ using positive allosteric modulators (PAMs) can treat most of the symptoms observed in preclinical models of schizophrenia. However, none of the PAMs developed so far have reached clinical trials, mainly due to adverse effects associated with intricate mGlu₅ signalling.

We recently solved cryoEM structures of mGlu₅ bound to several PAMs with different chemical structures and pharmacological properties, revealing the diversity of PAM binding modes within the mGlu₅ 7TM, and their impact on mGlu₅ receptor conformations. Based on this structural information, we searched for new PAMs with original structures, *in vitro* and *in vivo* biological activity. We performed an *in silico* screening of the Enamine Hit Locator Library and selected 100 compounds for testing *in vitro* biological activity, out of which PAM-27 induces IP₁ production and calcium mobilization. In order to further optimize the PAM-27 properties, we aimed at characterizing its binding mode at the allosteric binding site. We solved the PAM27-bound mGlu₅ receptor by cryoEM and refined its binding mode by using alanine mutant and molecular pharmacology. Finally, The PAM-27 displays *in vivo* activity in a

preclinical model of schizophrenia (*Disc-1 L100P* mutant) and offers a promising scaffold to develop molecules with antipsychotic properties.

Using these integrated approaches, we deliver a deep understanding of mGlu₅ allosteric modulation and allow the discovery of biased mGlu₅ PAMs with antipsychotic properties that are of high therapeutic potential.